

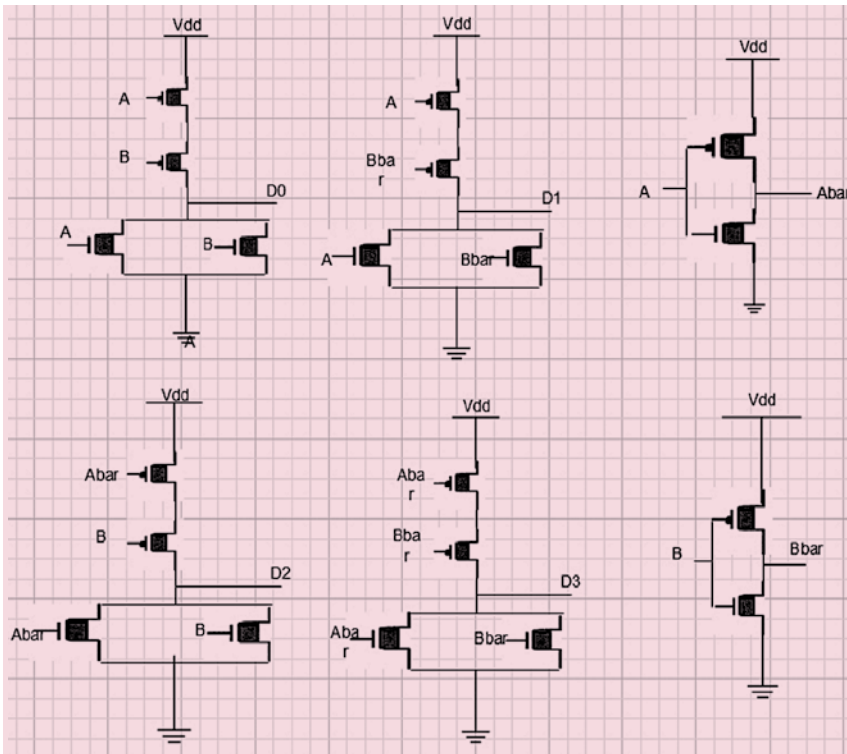


Fig. 5: CNTFET based 2:4 decoders (20T proposed)

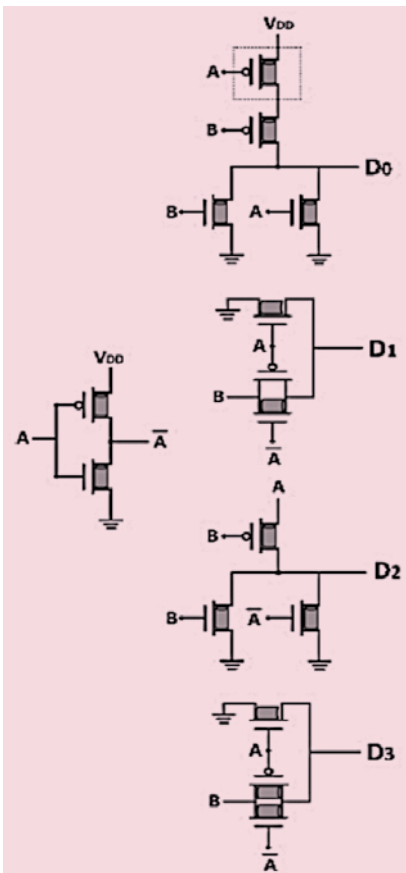


Fig. 6: Two-input 2 to 4 decoder using CNTFET 15T (proposed)

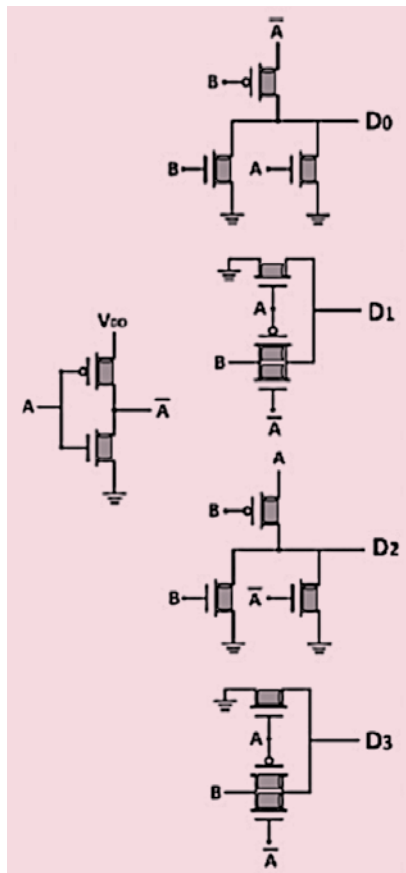


Fig. 7: Two-input 2 to 4 decoder using CNTFET (proposed)

input is applied to both the gates as well as to the source/drain diffusion terminals of the transistors. The logic uses either individual nMOS/pMOS pass transistors or parallel pairs of nMOS and pMOS known as transmission gates.

The AND/OR gates implemented by the transmission gate logic are efficient and thus can be extended in the application of decoder. Fig. 4 shows 2-input AND/OR gates implemented using TGL logic. The figure shows that AND/OR gates implemented using DVL and TGL logic require only three transistors.

Proposed models for CNTFET based decoders

Fig. 5 shows the proposed 2:4 decoder using CNTEFT based technology. There are 20 transistors used in the circuit. The PMOS are changed to PCNTFET and NMOS to NCNTFET, respectively.

15T mixed logic decoder using CNTFET. The proposed circuit in Fig. 6 is for 2 to 4 decoder using the 32nm CNTFET technology. In this only one inverter is used, eliminating the use of Bbar as the input and thus only 15 transistors are used to implement the circuit. The D0 part in the circuit is basically in CMOS logic and the rest in DVL/TGL mixed valued logics.

14T mixed logic decoder using CNTFET. Fig. 7 shows a two-input 2 to 4 decoder using CNTFET technology in 32nm. In this we mix TGL based AND/OR gates and DVL based AND/OR gates into one topology to form a 14T 2 to 4 decoder. The most important thing is that by combining we eliminate one inverter from the conventional 16T transistor topology, thereby reducing the total number of transistors as well as minimising the power dissipation.

We eliminated inverter B from the circuit and used DVL logic to implement D0 minterm in which A is used as a propagation signal and used TGL logic to implement D1 minterm in which logic B is used as a propagation signal. DVL logic is used for the minterm D2 in which A is the propagating signal and at last TGL is used to implement the midterm D3 where B is the propagation signal. As a result, the complementary B signal